TRUSTWORTHY STRESS DETECTION FROM WRIST-BASED PHYSIOLOGICAL SIGNALS

Group 4

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Proposal

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INTRODUCTION

RESEARCH PROBLEM

Physiological Arousal

- Stress
- Physical Activity
- Other Factors

→ Ambiguous Signals

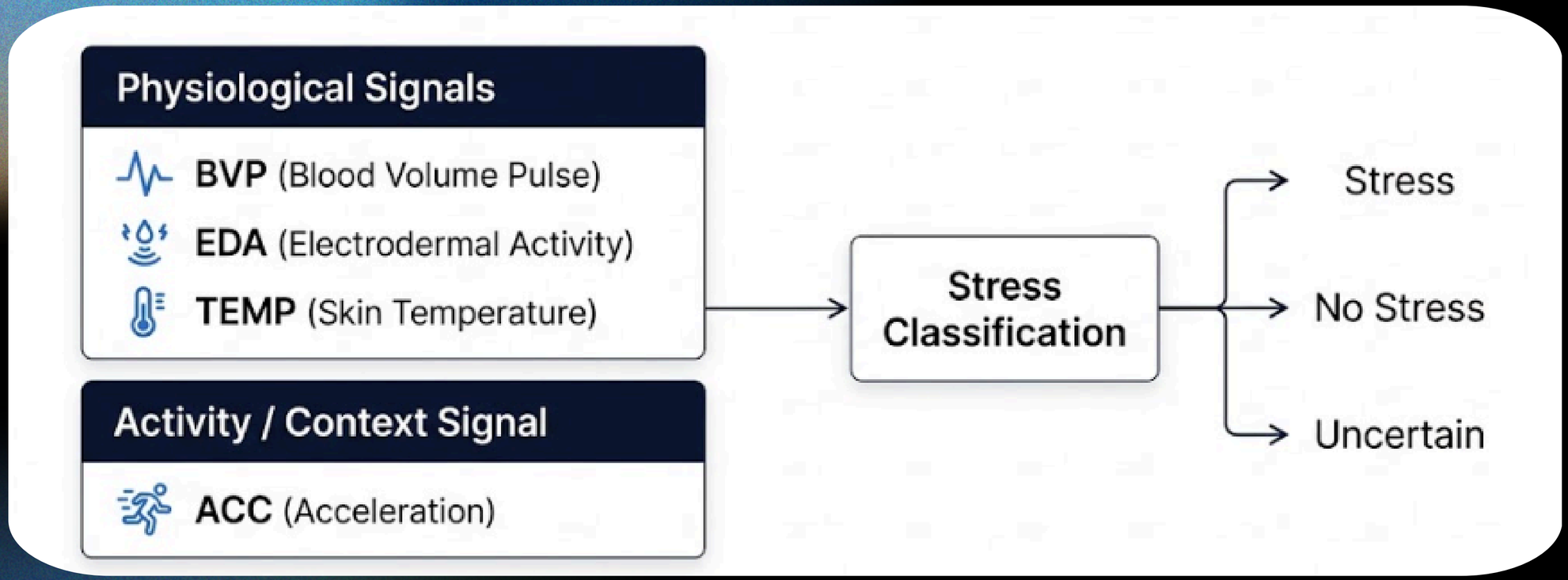
Is the prediction reliable enough?

Conventional Classifier
Stress / No Stress

DATASET: WESAD (Wearable Stress and Affect Detection)

- 15 participants
- Three states: **Neutral, Stress, Amusement**
- Chest-worn + wrist-worn sensors
- This project uses only the wrist-worn signals

INPUT - OUTPUT



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MODEL SELECTION

LOGISTIC REGRESSION	BASELINE & CALIBRATED PROBABILITIES
RANDOM FOREST	ROBUST TO NOISE & MOTION ARTIFACTS
XGBOOST	STATE-OF-THE-ART TABULAR PERFORMANCE

Feature-Based ML Rationale:

- Due to the small subject pool (N=15), deep learning poses a severe risk of subject-specific overfitting.
- We extract statistical, autonomic (HRV, EDA), and kinematic features into classical ML to ensure model interpretability and controllable probability calibration.

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EVALUATION FRAMEWORK

STANDARD VALIDATION

- **LOSO Cross-Validation:** Validates generalizability on unseen subjects; eliminates data leakage.
- **Macro F1 & PR-AUC:** Primary metrics to address severe class imbalance (Neutral vs. Stress).

SELECTIVE PREDICTION (CORE)

- **Coverage (%):** Fraction of non-abstained windows.
- **Selective F1:** F1-score restricted to accepted predictions.
- **ECE Metric:** Evaluates confidence-probability calibration.
- **FAR-Motion:** False alarm rate during high physical activity periods.

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BENCHMARKS & RESEARCH GAP

PUBLISHED BASELINES (WRIST-ONLY, LOSO):

Published Baselines (100% Coverage):

- **Schmidt et al. (2018) [Original]:** AdaBoost: 81.6% F1 | Random Forest: 79.2% F1
- **Bobade & Vani (2020) [ML/DL Survey]:** Ceiling for classical & deep models at ~84% F1
- **Critical Flaw:** Forced 100% prediction causes severe false alarms during wrist movement.

Our Proposed Target (Selective Decision):

- **Selective Strategy:** Abstains when signal entropy is high or motion is confounding.
- **Target Performance:** Selective F1 > 85.0%
- **Expected Coverage:** 80% – 90%
- **Key Trade-off:** Sacrificing ~15% ambiguous windows to eliminate false stress alarms.

WHY CONTEXT , UNCERTAINTY & ABSTENTION MATTER?

Problem: Physiological signals are not always stress-specific

EXAMPLE

Physical activity

↓

Heart Rate ↑ + EDA ↑

↓

Model may predict "Stress"

↓

False Stress Alert

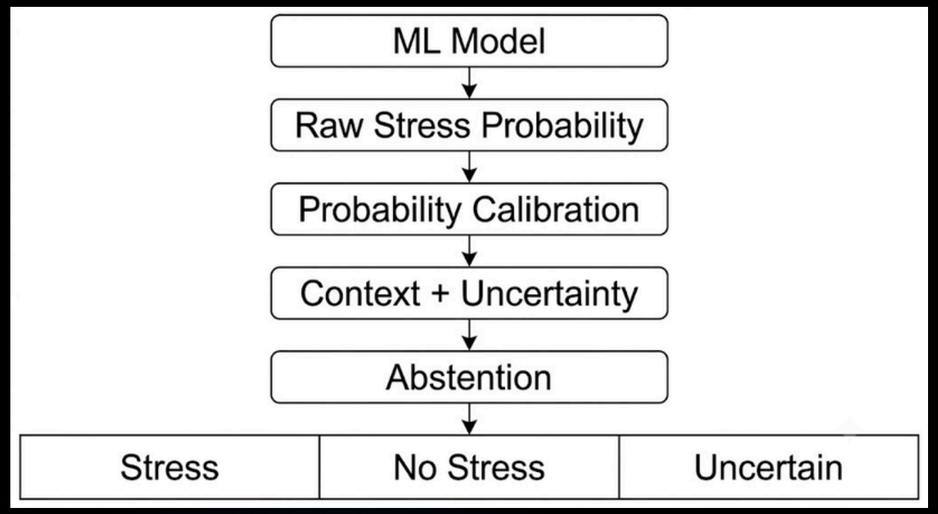
IDEA

- Use context information to distinguish stress from physical activity.
- Estimate prediction uncertainty before making a decision.
- Allow the model to abstain when the prediction is unreliable.

HOW DO WE HANDLE THIS ?

Probability Calibration & Abstention

PIPELINE BASIC



PROBABILITY CALIBRATION

Makes predicted probabilities better reflect actual likelihood.

Model: 0.80 Stress probability

↓

Calibration

↓

≈ 80% of similar predictions are actually Stress

ABSTENTION

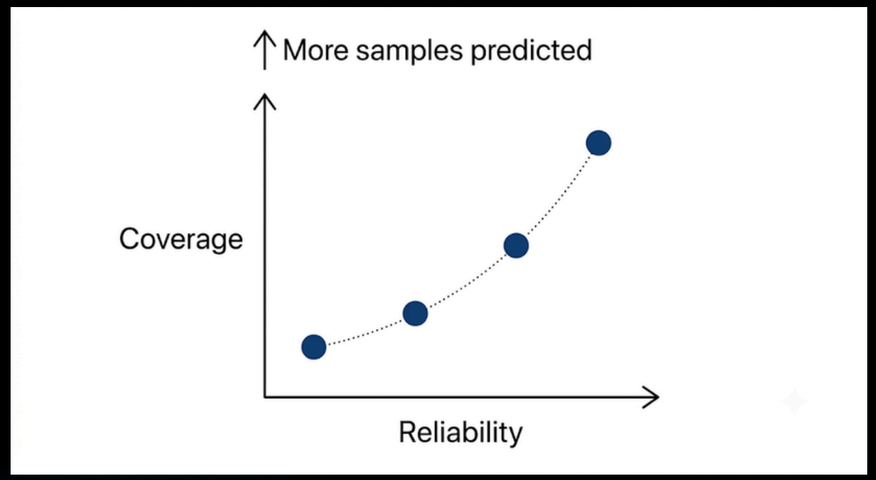
If confidence is insufficient, the model can refuse to make a definite prediction

High confidence → Stress / No Stress
Low confidence → Uncertain

HOW DO WE KNOW IT ACTUALLY HELPS?

Coverage–Reliability Trade-off

**How many predictions should the system make,
and how trustworthy should they be?**



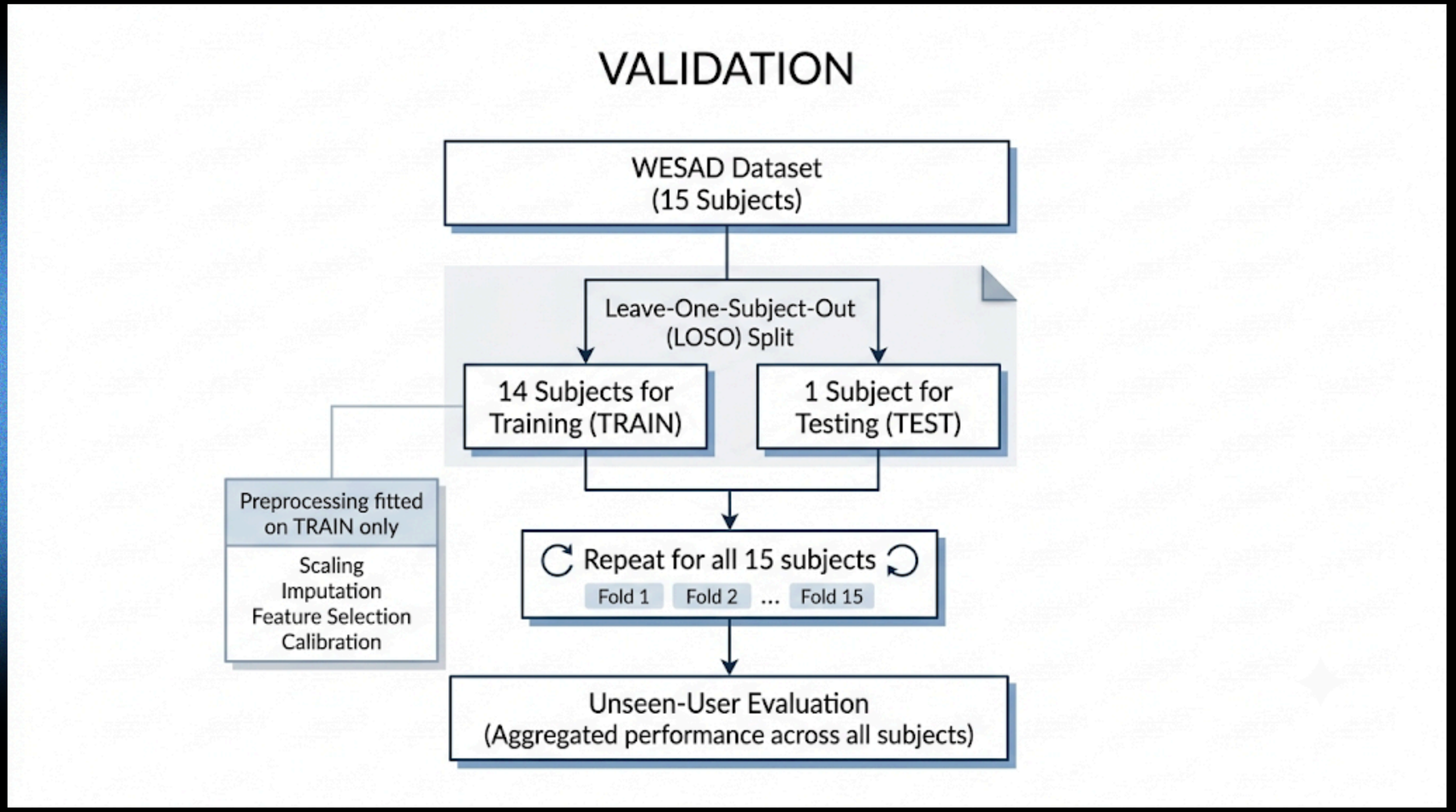
More Coverage	More Abstention
More predictions	Fewer predictions
More uncertain cases included	Uncertain cases removed
Potentially lower reliability	Potentially higher reliability

Find a suitable balance between coverage and reliability.

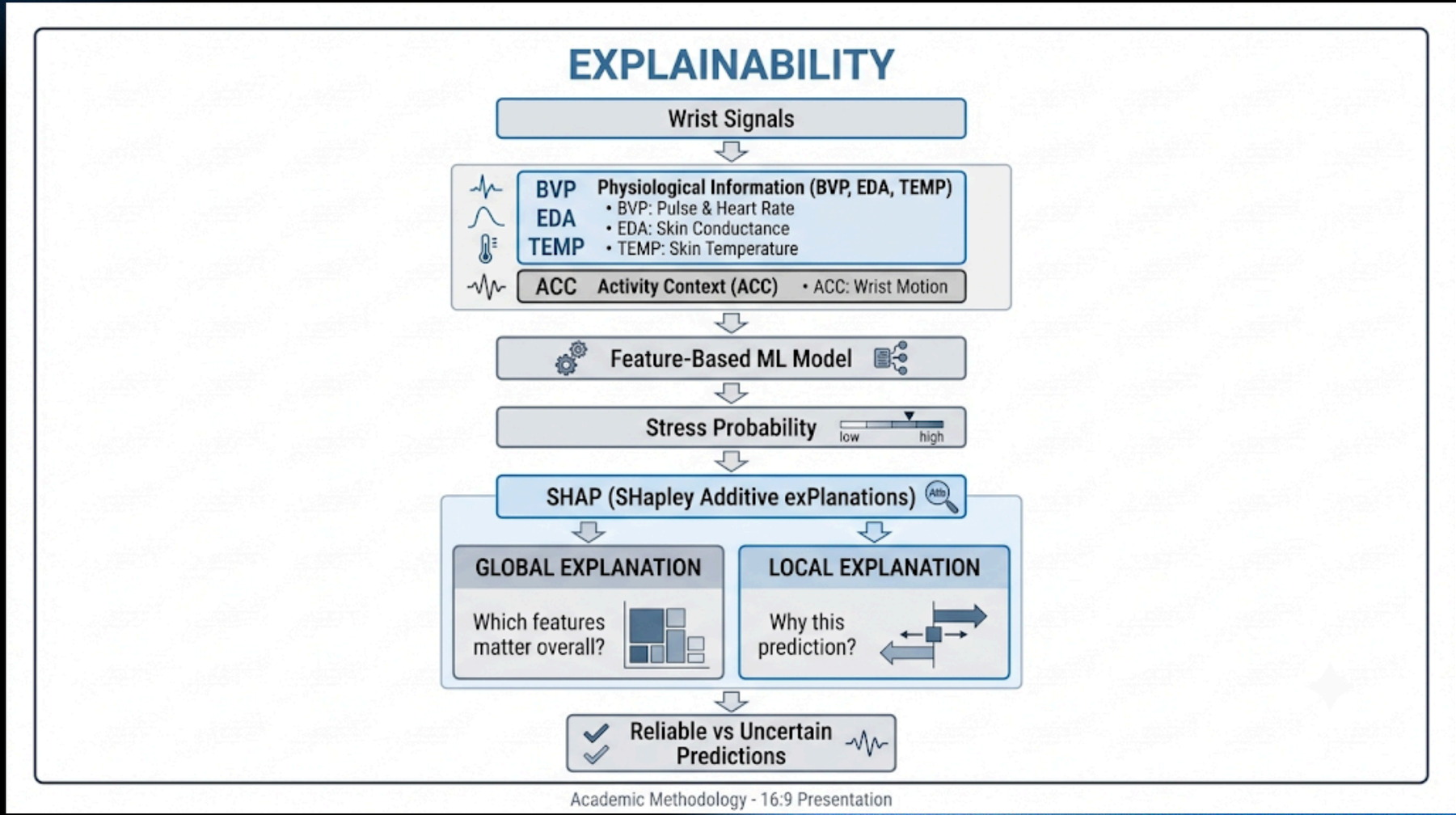
VALIDATION: GENERALIZATION TO UNSEEN USERS

Problem:

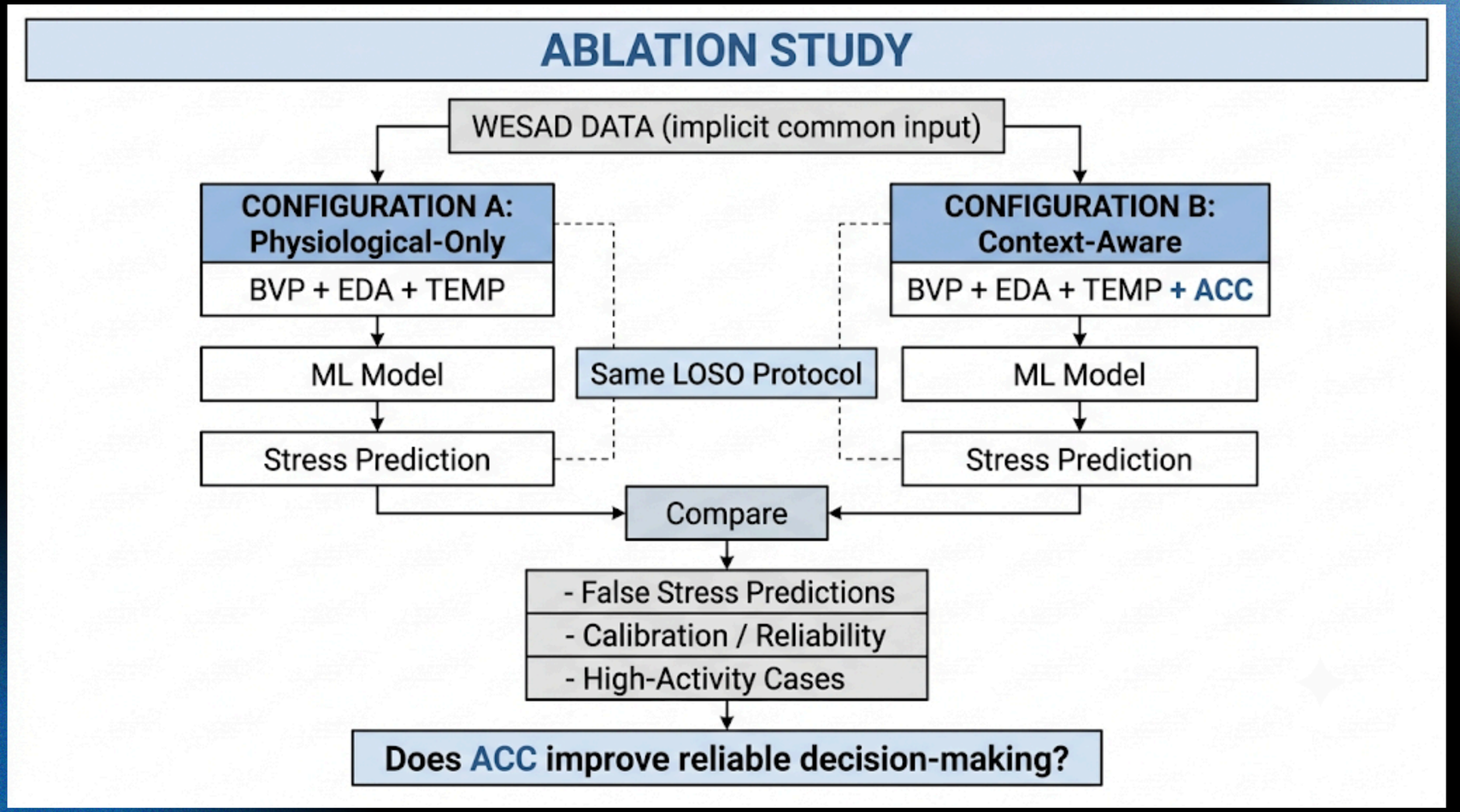
- Physiological signals are subject-specific.
- Randomly splitting windows may cause subject-level data leakage.



EXPLAINABILITY: WHY DOES THE MODEL PREDICT STRESS?



ABLATION STUDY: DOES ACTIVITY CONTEXT HELP?



THANK YOU